AIGR Gene Hypothesis Deep Research — GRIN1 (GluN1/NR1, Q05586)

Hypothesis slug: `homomeric-glun1-oocyte-currents-need-xenopus-glun2` **Focus term:** glutamate binding (GO:0016595) **Source:** `genes/human/GRIN1/GRIN1-ai-review.yaml` (free-text selector)

Summary

Executive verdict: PARTIALLY SUPPORTED. The curation-relevant core of the seed hypothesis is correct and strongly supported by primary evidence, but the specific mechanism the hypothesis proposes for the historical *Xenopus* oocyte observations is refuted (for the named XenU1 candidate) or unsupported (for a hypothetical "XenNR2B").

The **curation-critical claim** — that GluN1/NR1 (GRIN1, UniProt Q05586) **does not bind glutamate** and does not by itself form a functional glutamate-gated channel — is well established by three independent and complementary lines of primary evidence: (1) direct ligand binding, where glutamate binding to the isolated NR1 S1S2 ligand-binding core is "not detectable even at a 300 mM concentration" (PMID: 16641235); (2) two X-ray crystal structures that assign the glycine/D-serine site to NR1 (PMID: 12805203) and the glutamate site to NR2A (PMID: 16281028); and (3) clean mammalian (HEK293) heterologous expression, which lacks endogenous NMDAR subunits, where NR1 expressed alone is electrically silent unless an NR2 subunit is co-expressed (PMID: 16214956; PMID: 17105944; PMID: 11809857). This directly implies that the GO molecular-function annotation **glutamate binding (GO:0016595) is an over-annotation for GluN1**; the appropriate molecular function for GluN1 is **glycine binding (GO:0016594)**.

The **mechanistic sub-claim** — that the small glutamate/NMDA-activated, glycine-dependent currents historically recorded from *Xenopus laevis* oocytes injected with NR1 cRNA alone (e.g., the human NR1 reports PMID: 7685113 and PMID: 7679115) arise from heteromers between exogenous NR1 and an *endogenous Xenopus* GluN2-type subunit (named in the seed as "XenU1 / XenNR2B") — does **not** survive scrutiny. The proposed partner **XenU1 was experimentally excluded** by two independent primary studies (PMID: 11809857; PMID: 16214956), and XenU1

is in any case a **kainate-binding protein, not a GluN2-type subunit**. A targeted literature search returned **no primary identification** of an endogenous *Xenopus* oocyte "XenNR2B" GluN2 subunit that pairs with exogenous NR1. Thus, although NR1-alone currents are almost certainly **not** from homomeric GluN1 channels (consistent with the direct and structural data above), the **true origin of those oocyte currents remains unresolved**, and the seed's named mechanism should not be adopted as fact.

For a curator, the bottom line is clean and actionable: **remove / do not assign GO:0016595 (glutamate binding) to GluN1**, assign/retain **GO:0016594 (glycine binding)**, and treat the "endogenous Xenopus GluN2/XenNR2B partner" narrative as an unverified hypothesis rather than a citable mechanism.

Key Findings

Finding 1 — GluN1 (NR1) does not bind glutamate: direct biochemistry plus clean HEK293 functional assays

The most decisive evidence is direct and does not depend on the ambiguous oocyte system. Using a soluble, isolated NR1 S1S2 ligand-binding domain, glutamate binding to NR1 was **"not detectable even at a 300 mM concentration"** (PMID: 16641235), an extraordinarily high test concentration that effectively rules out any physiologically meaningful glutamate affinity. In the same body of work, glutamate binds the NR2 subunit and, weakly, the regulatory NR3A subunit — but not NR1.

This biochemical result is corroborated by functional reconstitution in HEK293 cells, a mammalian line that **lacks endogenous NMDAR subunits** and therefore provides a far cleaner test system than *Xenopus* oocytes. In HEK293, NR1 expressed alone does **not** produce a glutamate-gated response: "In HEK293 cells, we did not get receptor response unless an NR2 subunit was coexpressed" (PMID: 16214956). A second independent study states the point directly — "NR1 alone, which does not form functional channels" (PMID: 17105944). The field consensus architecture — a glycine-binding NR1 paired with a glutamate-binding NR2 — is reiterated across multiple recombinant studies (PMID: 17962328; PMID: 20407581).

Curation consequence: glutamate binding is a property of **GluN2**, not GluN1. GO:0016595 on GluN1 is unsupported by direct evidence.

Finding 2 — The specific XenU1 explanation for NR1-alone oocyte currents is refuted; the endogenous partner's identity is unresolved

The seed hypothesis names XenU1 (and a hypothetical "XenNR2B") as the endogenous *Xenopus* partner. Two independent primary studies directly tested, and rejected, the XenU1 hypothesis. Green et al. (2002) isolated XenU1 and XenU1a cDNAs and found that **co-injection of XenU1 with NR1 did not increase oocyte currents and did not form functional channels in HEK293**, with no pharmacological or biochemical evidence of association — concluding that "**XenU1 does not associate with the NR1 subunit and that an alternative explanation must be sought for the channels observed when NR1 is expressed alone in oocytes**" (PMID: 11809857). Schmidt, Werner & Hollmann (2006) revisited the postulated "unitary glutamate receptor," cloned *Xenopus* NR1 variants, and showed that oocyte glutamate/glycine responses "**were independent of coexpressed XenU1**" (PMID: 16214956).

Critically, **XenU1 is a kainate-binding protein, not a GluN2-type subunit**, so even its name in the seed hypothesis conflates two distinct molecular entities. A targeted search for a primary report of an endogenous *Xenopus* oocyte GluN2-type subunit ("XenNR2B") that pairs with exogenous NR1 returned **no primary identification**. The seed's mechanistic claim is therefore refuted in its specific (XenU1) form and merely asserted — without primary support — in its generalized ("XenNR2B") form.

Finding 3 — Crystal structure: the GluN1 ligand-binding core forms the glycine/D-serine site, not a glutamate site

Furukawa & Gouaux (2003) solved co-crystal structures of the NR1 S1S2 ligand-binding core bound to glycine, D-serine, the partial agonist D-cycloserine, and the glycine-site antagonist 5,7-dichlorokynurenic acid. The bilobed "clamshell" domain closes on glycine-site agonists and opens with the antagonist, and the authors state that "**NMDA receptors require both glycine and glutamate for activation with NR1 and NR2 forming glycine and glutamate sites, respectively**" (PMID: 12805203). This structurally assigns the **glycine** site to NR1 and complements the direct binding data (Finding 1).

Finding 4 — The glutamate-binding site is structurally localized to GluN2 (NR2A), not GluN1

Furukawa, Singh, Mancusso & Gouaux (2005) solved the crystal structure of the NR2A ligand-binding core bound to glutamate and the NR1–NR2A heterodimer with glutamate plus glycine. The NR2A–glutamate complex "**defines the determinants of glutamate and NMDA**

recognition," confirming the NR1–NR2A heterodimer as the functional ligand-binding unit (PMID: 16281028). This directly localizes glutamate binding (GO:0016595) to **GluN2**, completing the structural case: the glutamate site is physically on NR2, not NR1.

Mechanistic Model / Interpretation

The four findings converge on a single, internally consistent model of NMDA-receptor ligand-site architecture and explain why the historical oocyte observations were misleading.

```
          NMDA receptor ligand-binding architecture
          -----------------------------------------

GluN1 (NR1, GRIN1, Q05586)            GluN2 (NR2A-D)
--------------------------            --------------------
S1S2 clamshell binds:                 S1S2 clamshell binds:
  glycine / D-serine                    GLUTAMATE / NMDA
  (co-agonist site)                      (agonist site)
GO:0016594 glycine binding  <--      --> GO:0016595 glutamate binding
Does NOT bind glutamate                  Does NOT bind glycine
(<none to 300 mM; <a href="https://pubmed.ncbi.nlm.nih.gov/16641235/" rel="noopener noreferre

Functional channel = (GluN1)2 : (GluN2)2 heterotetramer
Requires BOTH co-agonists; NR1 alone is non-functional.
```

Why did NR1-alone oocyte currents ever appear (as in PMID: 7685113 and PMID: 7679115)? The seed's instinct is partially right: these currents are almost certainly **not** from homomeric GluN1 channels, because (a) the isolated NR1 domain cannot bind glutamate at all, and (b) in a clean mammalian system without endogenous subunits, NR1 alone is silent. The remaining question — what the exogenous NR1 assembles *with* in the oocyte — was the explicit subject of two follow-up studies, and the leading candidate (XenU1) was excluded. So the model is:

Claim	Status	Basis
NR1-alone oocyte currents are from **homomeric GluN1**	**Refuted**	Direct binding (16641235), clean HEK293 silence (16214956, 17105944), structure (12805203, 16281028)
The currents require an **endogenous Xenopus partner**	Plausible but **unproven**	Inference from NR1 non-functionality; no partner positively identified
The partner is **XenU1**	**Refuted**	Direct tests (11809857, 16214956); XenU1 is a kainate-binding protein
The partner is an endogenous **"XenNR2B"** GluN2	**Unsupported**	No primary report found
GluN1 **binds glutamate** (GO:0016595)	**Refuted / over-annotation**	All of the above
GluN1 **binds glycine** (GO:0016594)	**Supported**	Structure (12805203), consensus (17962328, 20407581)

The practical upshot for GO curation is that the over-annotation and the correct annotation can both be resolved independently of ever settling what the oocyte partner was: the direct biochemistry and structures are sufficient.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confid limitat
PMID: 16641235	Direct assay (ligand binding)	**Supports** core (refutes glutamate-binding)	Does NR1 LBD bind glutamate?	Glutamate binding to NR1 "not detectable even at a 300 mM concentration"	Soluble NR1 S1S2 domain, in vitro	High; isolate domai but concer tested decisiv
PMID: 12805203	Structural	**Supports** (NR1 = glycine site)	Which ligand site does NR1 form?	NR1 S1S2 clamshell binds glycine/ D-serine; "NR1 and NR2 forming glycine and glutamate sites, respectively"	X-ray crystallography, rat NR1 LBD	High; refere contex
PMID: 16281028	Structural	**Supports** (glutamate site = NR2)	Where is the glutamate site?	NR2A LBD + glutamate structure; NR1–NR2A heterodimer defines glutamate/ NMDA recognition	X-ray crystallography	High
PMID: 16214956	Direct assay + heterologous	**Supports** core; **refutes** XenU1 mechanism	Is NR1 functional alone? Does XenU1 mediate oocyte currents?	No HEK293 response without NR2; oocyte responses independent of coexpressed	HEK293 + Xenopus oocytes	High; c addres seed

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confid limitat
				XenU1; no "unitary glutamate receptor"		
PMID: 11809857	Interaction/ functional	**Refutes** XenU1 mechanism	Does XenU1 associate with NR1?	XenU1 + NR1 gave no increased current, no HEK293 channels; "XenU1 does not associate with NR1"	Xenopus oocytes + HEK293	High; definit XenU1
PMID: 17105944	Direct assay (heterologous)	**Supports** core	Is NR1 alone functional in HEK293?	"NR1 alone, which does not form functional channels"	HEK293 (NR1/ NR2B toxicity study)	High f stated
PMID: 7685113	Direct assay (electrophysiology)	**Competing / context**	Can NR1 cRNA alone yield NMDA currents?	Human NR1 cRNA in oocytes gave glutamate/ NMDA-activated, glycine-dependent currents (interpreted then as homomeric)	Xenopus oocytes, human hNR1	Histor source artifac interp supers
PMID: 7679115	Direct assay (electrophysiology)	**Competing / context**	Same as above	Single cloned human NMDAR1 protein in	Xenopus oocytes, human NMDAR1	Histor source interp supers

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confid limitat
				oocytes shows NMDAR-like properties		
PMID: 17962328	Consensus + assay	**Supports** (architecture)	Co-agonist subunit assignment	"coagonist glycine-binding NR1 subunits and glutamate-binding NR2 subunits"	Recombinant NMDARs, oocytes	High f archite statem
PMID: 20407581	Assay + review	**Qualifies**	Glutamate-independent NR1/NR3 receptors	NR1/NR3 lack glutamate sites, gated by glycine alone; classical NMDARs = glycine-NR1 + glutamate-NR2	Oocytes	Qualifi NR1 fu withou glutam in NR3 contex

GO Curation Implications

Lead (requires curator verification): The molecular-function term **glutamate binding (GO:0016595) should NOT be annotated to GluN1/GRIN1 (Q05586)**. If it is currently present, it is an over-annotation and should be **removed or down-qualified**; if proposed, it should be **rejected**.

- **MF — remove/reject:** GO:0016595 *glutamate binding* for GluN1. Direct binding (P 16641235), structure (P 12805203

), P 16281028 and clean functional assays (P 16214956) P 17105944 place the glutamate site on GluN2.

- **MF — assign/retain (more informative replacement): GO:0016594 *glycine binding*** for GluN1, supported structurally (P 12805203) and by consensus (P 17962328). P 20407581 This is the correct, specific molecular function and directly replaces the incorrect glutamate-binding term. (Avoids the uninformative "protein binding" fallback.)
- **MF/CC context:** GluN1 is an obligatory subunit of the heterotetrameric NMDA receptor cation channel; GO terms for *glutamate-gated calcium-permeable cation channel activity* and *NMDA receptor complex* (CC) are appropriate for the **complex-level** function, but the glutamate *binding* activity itself belongs to the GluN2 subunits, not GluN1.
- **Provenance caveat:** The historical primary papers that reported NR1-alone oocyte currents (PMID: 7685113, PMID: 7679115) are a likely **database carry-over source** for any glutamate-binding annotation on GluN1. Curators should check whether an existing GO:0016595 annotation traces (via IDA/IMP on these oocyte papers) to that superseded interpretation.

Mechanistic Scope

The immediate molecular function under test is **ligand (glutamate) binding by the GluN1 gene product itself**. The evidence cleanly separates:

- **Direct gene-product activity (GluN1):** glycine/D-serine co-agonist binding (the NR1 S1S2 clamshell) and an obligatory structural/gating role in the heterotetramer. GluN1 does **not** directly bind glutamate.
- **Complex-level / downstream properties:** glutamate-gated cation flux, Ca^{2+} permeability, Mg^{2+} block, and channel gating are **emergent properties of the assembled GluN1/GluN2 heterotetramer**, not of GluN1 alone. The oocyte "NR1-alone" currents are a downstream, system-dependent artifact of expressing NR1 in a cell that may supply an unidentified endogenous partner — not evidence of a GluN1 intrinsic glutamate-binding activity.

Attributing glutamate binding to GluN1 conflates a **complex-level phenotype** (the channel opens to glutamate) with a **subunit-level molecular activity** (which subunit's pocket the glutamate occupies). GO MF annotations should capture the latter.

Conflicts and Alternatives

1. **Historical primary data vs. modern biochemistry.** The 1993 oocyte papers (PMID: 7685113, PMID: 7679115) genuinely recorded glutamate/NMDA-activated, glycine-dependent currents from NR1 cRNA alone and interpreted them as homomeric NR1 receptors. This is the principal conflicting evidence, but the interpretation is superseded: the currents are real, their *homomeric* attribution is not.
2. **Paralog/subunit confusion.** The seed's "glutamate binding" belongs to the **paralogous GluN2** subunits, not GluN1 — a classic subunit-level over-annotation.
3. **Entity confusion in the seed itself.** "XenU1" (a kainate-binding protein) and "XenNR2B" (a hypothetical GluN2) are different molecular classes; the seed treats them as interchangeable candidates. Only XenU1 has been tested (and excluded); "XenNR2B" has no primary report.
4. **Organism-specific expression system.** *Xenopus* oocytes are notorious for endogenous channels/partners; HEK293 (lacking endogenous NMDAR subunits) is the appropriate counter-system and shows NR1 silence — isolating the oocyte result as system-specific.

5. **Isoform qualifier (NR3 context).** NR1/NR3 receptors are glutamate-independent, glycine-gated "excitatory glycine receptors" (PMID: 20407581) — reinforcing that GluN1's intrinsic ligand site is for glycine, not glutamate.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
Identity of the endogenous *Xenopus* oocyte partner for exogenous NR1	Targeted PubMed searches for "XenNR2B"/ endogenous Xenopus GluN2 pairing with NR1; XenU1 excluded (P 11809857 P 16214956)	Determines whether the seed's *mechanism* can be cited at all; currently it cannot	Pull-down/MS of NR1 complexes from injected oocytes; RNA-seq of oocyte NMDAR-family transcripts; knockdown of candidate endogenous subunits
Current GO annotation status of GO:0016595 on GluN1 in the live GO/UniProt record	Not directly queried in this run (reasoned from primary evidence)	The recommended action (remove vs. reject) depends on whether the term is already present and its evidence code	Query QuickGO/UniProt for Q05586 MF annotations and their evidence/reference IDs
Whether any GluN1 glutamate-binding annotation traces to P 7685113 / 7679115	Inferred as likely carry-over	Identifies the citation to correct	Trace annotation provenance in GO annotation file (GAF)
Quantitative affinity of glutamate for NR3A vs. NR1	Noted qualitatively (glutamate binds NR3A weakly; not NR1)	NR3-containing receptors are an exception to "glutamate on NR2"	Review P 16641235 primary binding constants

Additional limitations: This run relied on literature evidence and reasoning; it did not programmatically query the live GO/QuickGO/UniProt annotation record for Q05586, so the precise current annotation status of GO:0016595 on GluN1 is **inferred, not verified**. Some

supporting statements (P 12805203) (P 16281028) were taken from provided abstract snippets/reference context rather than re-read full texts in this run. The identity of the endogenous *Xenopus* oocyte partner remains genuinely unknown; no claim is made beyond "not XenU1, and no XenNR2B identified."

Discriminating Tests / Proposed Follow-up Actions

1. **Trace the annotation provenance (highest-value curator action).** Query QuickGO/UniProt for Q05586 and determine whether GO:0016595 is present, its evidence code, and its `with/from` reference. This single check decides remove-vs-reject.
2. **Confirm the direct-binding and structural primaries.** Verify the exact quotes in

   P 16641235

   (300 mM glutamate undetectable),

   P 12805203

   (NR1 = glycine site), and

   P 16281028

   (NR2A = glutamate site) against the full texts.
3. **(If the mechanism must be resolved)** Affinity-purify NR1 complexes from NR1-injected *Xenopus* oocytes and identify co-assembled endogenous subunits by mass spectrometry; complement with oocyte NMDAR-family transcriptome profiling. This would settle the unresolved partner question that the seed guessed at.
4. **Cross-system control.** Reaffirm NR1-alone silence in HEK293 and (ideally) in a second mammalian line to cement that the oocyte current is system-specific.

Curation Leads (require curator verification)

- **Action change:** Remove / do not assign **GO:0016595 (glutamate binding)** to GluN1 (GRIN1, Q05586). Treat as over-annotation at the subunit level.
- **Replacement MF term:** Assign/retain **GO:0016594 (glycine binding)** as GluN1's specific, evidence-supported molecular function.
- **Candidate references with snippets to verify:**
- PMID: 16641235 — "binding of glutamate to NR1 was not detectable even at a 300 mm concentration" (direct refutation of glutamate binding).
- PMID: 12805203 — "NR1 and NR2 forming glycine and glutamate sites, respectively" (glycine site on NR1).
- PMID: 16281028 — NR2A LBD + glutamate structure "defines the determinants of glutamate and NMDA recognition" (glutamate site on NR2).
- PMID: 16214956 — "we did not get receptor response unless an NR2 subunit was coexpressed"; oocyte responses "independent of coexpressed XenU1."
- PMID: 11809857 — "XenU1 does not associate with the NR1 subunit."
- PMID: 17105944 — "NR1 alone, which does not form functional channels."
- **Candidate note for the review's free-text field:** The seed's claim that NR1-alone oocyte currents "come from heteromeric receptors formed between injected GluN1 and an endogenous *Xenopus* GluN2-type subunit" should be flagged as **mechanistically unverified** — XenU1 was tested and excluded, and no "XenNR2B" has been identified in primary literature. The *conclusion* (GluN1 does not bind glutamate / is not functional alone) is correct; the *stated mechanism* is not citable.
- **Suggested curator questions:** (a) Is GO:0016595 currently annotated to Q05586, and via which reference? (b) Does that annotation descend from the 1993 oocyte papers? (c) Is GO:0016594 already present?
- **Suggested experiments (for the record):** MS identification of the oocyte endogenous partner; oocyte NMDAR-family transcriptomics.

Evidence Base (literature summary)

- **Reference-context primaries:** PMID: 7685113 and PMID: 7679115 are the historical human-NR1 oocyte reports whose "homomeric" interpretation the hypothesis (correctly) challenges; PMID: 12805203 is the NR1 glycine-site crystal structure.
- **Decisive refutation of glutamate binding:** PMID: 16641235 (direct binding) and PMID: 16281028 (glutamate site on NR2A).
- **Clean-system functional confirmation:** PMID: 16214956, PMID: 17105944.
- **XenU1 mechanism refutation:** PMID: 11809857, PMID: 16214956.
- **Architecture/consensus:** PMID: 17962328, PMID: 20407581.

Conclusion

The hypothesis is **partially supported**: its curation-relevant conclusion (GluN1 does not bind glutamate and is not a functional glutamate-gated channel alone → GO:0016595 is an over-annotation for GluN1; GO:0016594 glycine binding is correct) is strongly supported by direct biochemistry, two crystal structures, and clean HEK293 assays. Its specific mechanistic explanation for the oocyte currents (an endogenous *Xenopus* GluN2 / XenU1 / "XenNR2B" partner) is refuted for XenU1 and unsupported for "XenNR2B," and should be treated as an unverified hypothesis rather than citable mechanism.
